

- A. Electronic devices that perform only one operation can be programmable device.
- B. Logic devices provide timing and control operations function
- C. Internet is example of network which connect millions of people
- D. None

4. Which of the following file type has self-describing structure?

- A. XML
- B. SQL databases
- C. PDF file
- D. NO-SQL database

5. Which of the following characteristics of big data indicates data in uncertainty due to data inconsistency and ambiguity?

- A. Velocity
- B. Volume
- C. Variety
- D. veracity

6. _____ is performed by expert annotators that are responsible for improving the accessibility and quality of data.

- A. Data Acquisition
- B. Data Analysis
- C. Data Curation
- D. Data Storage

7. _____ is a framework that allows for the distributed processing of large datasets across clusters of computers using simple programming models.

- A. Hadoop
- B. NoSQL
- C. Big data
- D. API
- E. None

8. Which technique in big data life cycle is used to transfer data from various sources to Hadoop?

- A. Processing
- B. Ingesting
- C. Analysing
- D. Visualizing

9. Which one of the following is **not** true about cluster computing?

- A. It is economical
- B. It is scalable
- C. It is flexible
- D. It is rigid

10. **Chatbots and Roboadvisors** are best example for.

- A. Context aware and Retention
- B. Domain – Specific Expertise
- C. Rule based Systems
- D. Reasoning Machines

11. Which one is **NOT** application of AI?

- A. Recommendation
- B. Snap chat
- C. Email spam filters
- D. Voice recorder

12. Which of the following statement is **not** true?

3. What are the types of AI based on their functionality? Explain briefly (3 pt.).

1. ~~Short~~ ^{limited} memory - its memory save for short past experiences
2. ~~limited~~ ^{short} memory - doesn't save past experiences
3. Theory of mind - doing tasks as human

4. What are the goals for developing Artificial Intelligence? (2 pt.)

- ~~to build machine~~ Building machine
- Replication of human intelligence
- an intelligent connection of perception action
- creating system which can perform as human.

- A. The most common and currently available AI is Narrow AI
- B. The most common uses of AI today fit Rule-Based Systems level
- ☒ C. General AI is still under research
- D. self-driving car is an example of General AI

13. Which one is different?

- A. Super intelligence
- B. High accuracy with less failure
- C. Useful for risky area
- D. Digital assistance

14. Which of the following is **not** an example of reactive machine?

- A. Deep Blue
- B. AlphaGo
- C. Self-driving cars
- D. None

15. In mapping human thinking to AI components, which layer is related to taking action or making decisions;

- A. Sensing
- B. Interacting
- C. Interpreting
- D. Observing

Part III: Write precise and clear answers for the following questions.

1) Write at least four advanced technologies of fourth industrial revolution? (2 pt.)

- Self-driving cars

- ASI

- AGI

2) Briefly explain data types from data analytics perspective; provide examples for each of them. (3pts.)

1. ~~Structured~~ Structured - data that is structured by row & columns eg. Excel & XML

2. Semistructured - it doesn't process fully eg. No-SQL

3. Unstructured - it is raw that wants to input eg. Audio files.